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BATTERY MODULE

Abstract

A battery module includes: a first battery cell; a first case to accommodate the first battery cell, and having a first through region to expose a portion of a surface of the first battery cell therethrough; a second case to accommodate at least a portion of the first case; and a first heat conducting member between the first battery cell and the second case.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Application No. 10-2024-0030982, filed on Mar. 4, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference herein.

BACKGROUND

1. Field

[0002] Aspects of embodiments of the present disclosure relate to a battery module having improved heat dissipation performance.

2. Description of the Related Art

[0003] Unlike primary batteries that are not designed to be (re)charged, secondary (or rechargeable) batteries are batteries that are designed to be discharged and recharged. Low-capacity secondary batteries are used in portable, small electronic devices, such as smart phones, feature phones, notebook computers, digital cameras, and camcorders, while large-capacity secondary batteries are widely used as power sources for driving motors in hybrid vehicles and electric vehicles and for storing power (e.g., home and/or utility scale power storage). A secondary battery generally includes an electrode assembly composed of a positive electrode and a negative electrode, a case accommodating the same, and electrode terminals connected to the electrode assembly.

[0004] The secondary battery may generate heat during discharge and/or charge. In a case where the heating persists, the secondary battery may undergo a thermal runaway, which may cause a fire in a device or a vehicle equipped with the secondary battery.

[0005] The above information disclosed in this Background section is for enhancement of understanding of the background of the present disclosure, and therefore, it may contain information that does not constitute related (or prior) art.

SUMMARY

[0006] One or more embodiments of the present disclosure may be directed to a battery module having improved heat dissipation performance.

[0007] These and other aspects and features of the present disclosure will be described in or will be apparent from the following description of embodiments of the present disclosure.

[0008] According to one or more embodiments of the present disclosure, a battery module includes: a first battery cell; a first case configured to accommodate the first battery cell, and having a first through region to expose a portion of a surface of the first battery cell therethrough; a second case configured to accommodate at least a portion of the first case; and a first heat conducting member between the first battery cell and the second case.

[0009] In an embodiment, a first surface of the first heat conducting member may be in direct contact with a portion of a surface of the first battery cell exposed through the first through region, and a second surface of the first heat conducting member may be in contact with the second case.

[0010] In an embodiment, the first heat conducting member may be located in the first through region of the first case.

[0011] In an embodiment, an area of the first through region may be smaller than an area of a first surface of the first battery cell.

[0012] In an embodiment, a length of a long side of the first through region may be shorter than a length of a long side of a first surface of the first battery cell, and a length of a short side of the first through region may be shorter than a length of a short side of the first surface of the first battery cell.

[0013] In an embodiment, the first case may be configured to accommodate a plurality of battery

cells including the first battery cell, the first case may have a plurality of through regions including the first through region, each of the through regions may correspond to a single battery cell from among the plurality of battery cells, and a plurality of heat conducting members including the first heat conducting member may be located in the through regions, respectively.

[0014] In an embodiment, the first case may be configured to accommodate a plurality of battery cells including the first battery cell, the first case may have a plurality of through regions including the first through region, each of the through regions may correspond to two or more battery cells from among the plurality of battery cells, and a plurality of heat conducting members including the first heat conducting member may be located in the through regions, respectively.

[0015] In an embodiment, a thickness of the first heat conducting member may be thicker than a thickness of the first case having the first through region.

[0016] In an embodiment, the first case may include an insulating material.

[0017] In an embodiment, the second case may include a metal.

[0018] In an embodiment, the battery module may further include: a second battery cell; and a second heat conducting member between the second battery cell and the second case. The second case may have a second through region to expose a portion of a surface of the second battery cell therethrough.

[0019] According to one or more embodiments of the present disclosure, a method of assembling a battery module, includes: preparing a first case for accommodating a battery cell, and having a through region to expose a portion of a surface of the battery cell therethrough; preparing a second case having an internal space for accommodating at least a portion of the first case; coupling the first case and the second case together so that the first case is seated in the internal space of the second case; disposing a heat conducting member in the through region of the first case; and seating the battery cell in the first case to be disposed on the through region.

[0020] In an embodiment, in the seating of the battery cell in the first case, a first surface of the heat conducting member may be in direct contact with a portion of a surface of the battery cell exposed through the through region, and a second surface of the heat conducting member may be in contact with the second case.

[0021] In an embodiment, in the preparing of the first case, the through region may be formed in the first case so that an area of the through region may be smaller than an area of a first surface of the battery cell.

[0022] In an embodiment, in the preparing of the first case, the through region may be formed in the first case so that a length of a long side of the through region may be shorter than a length of a long side of a first surface of the battery cell, and a length of a short side of the through region may be shorter than a length of a short side of the first surface of the battery cell.

[0023] In an embodiment, the first case may be configured to accommodate a plurality of battery cells including the battery cell, the first case may have a plurality of through regions including the through region, each of the through regions may correspond to a single battery cell from among the plurality of battery cells, and a plurality of heat conducting members including the heat conducting member may be disposed in the through regions, respectively.

[0024] In an embodiment, the first case may be configured to accommodate a plurality of battery cells including the battery cell, the first case may have a plurality of through regions including the through region, each of the through regions may correspond to two or more battery cells from among the plurality of battery cells, and a plurality of heat conducting members including the heat conducting member may be disposed in the through regions, respectively.

[0025] In an embodiment, a thickness of the heat conducting member may be thicker than a thickness of the first case having the through region.

[0026] In an embodiment, the first case may include an insulating material.

[0027] In an embodiment, the second case may include a metal.

[0028] According to some embodiments of the present disclosure, the heat conducting member may

be disposed between a battery cell and the metal case, and heat generated by the battery cell may be dissipated to the outside through the heat conducting member and the case.

[0029] According to some embodiments of the present disclosure, with the through regions being provided in the casing and the heat conducting members being disposed in the through regions, a larger area of the heat conducting members may be in contact with the battery cells. With the increase in the area where the battery cells are in contact with the heat conducting members, the heat dissipation effect of the battery cells may be improved.

[0030] According to some embodiments of the present disclosure, the case may be used as a heat sink, thereby increasing the heat capacity.

[0031] According to some embodiments of the present disclosure, as the battery cells are seated in the internal space of a combined structure of the first case associated with insulation and the second case associated with heat dissipation, both insulation and heat dissipation for the battery cells may be achieved.

[0032] However, aspects and features of the present disclosure are not limited to those described above, and other aspects and features not mentioned will be clearly understood by a person skilled in the art from the detailed description, described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] The following drawings attached to this specification illustrate embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed as being limited to the drawings:

[0034] FIG. 1 illustrates an exploded perspective diagram showing a battery module according to one or more embodiments of the present disclosure;

[0035] FIG. 2 illustrates a perspective view showing a secondary battery according to one or more embodiments of the present disclosure;

[0036] FIG. 3 is a cross-sectional view taken along the line II-II in FIG. 2;

[0037] FIG. 4 illustrates a perspective view showing a first case and a second case according to one or more embodiments of the present disclosure;

[0038] FIG. 5 illustrates a perspective diagram showing a coupled state of the first case and the second case according to one or more embodiments of the present disclosure;

[0039] FIG. 6 illustrates a plan diagram showing the coupled state of the first case and the second case according to one or more embodiments of the present disclosure;

[0040] FIG. 7 illustrates a state where heat conducting members are disposed in a plurality of through regions according to one or more embodiments of the present disclosure;

[0041] FIG. 8 illustrates an example of a battery module in which a plurality of battery cells are inserted in the first case in a state where the first case and the second case are coupled together according to one or more embodiments of the present disclosure;

[0042] FIG. 9 illustrates through regions having various shapes according to one or more embodiments of the present disclosure;

[0043] FIG. 10 illustrates a battery pack according to one or more embodiments of the present disclosure;

[0044] FIG. 11 illustrates a battery pack according to one or more embodiments of the present disclosure;

[0045] FIG. 12 illustrates a vehicle body and vehicle body parts having a battery pack according to one or more embodiments of the present disclosure;

[0046] FIG. 13 illustrates a vehicle body and vehicle body parts having a battery pack according to

one or more embodiments of the present disclosure; and [0047] FIG. 14 illustrates a flowchart showing a method of assembling a secondary battery module according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0048] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in this specification and claims should not be construed as being limited to the usual or dictionary meaning and should be interpreted as meaning and concept consistent with the technical idea of the present disclosure based on the principle that the inventor can be his/her own lexicographer to appropriately define the concept of the term to explain his/her invention in the best way.

[0049] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical ideas, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0050] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.

[0051] In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B and C,” “at least one of A, B or C,” “at least one selected from a group of A, B and C,” or “at least one selected from among A, B and C” are used to designate a list of elements A, B and C, the phrase may refer to any and all suitable combinations or a subset of A, B and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0052] It will be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0053] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned

over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated **90** degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

[0054] The terminology used herein is for the purpose of describing embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0055] Also, any numerical range disclosed and/or recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein. All such ranges are intended to be inherently described in this specification such that amending to expressly recite any such subranges would comply with the requirements of 35 U.S.C. § 112(a) and 35 U.S.C. § 132(a).

[0056] References to two compared elements, features, etc. as being “the same” may mean that they are “substantially the same”. Thus, the phrase “substantially the same” may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.

[0057] Throughout the specification, unless otherwise stated, each element may be singular or plural.

[0058] Arranging an arbitrary element “above (or below)” or “on (under)” another element may mean that the arbitrary element may be disposed in contact with the upper (or lower) surface of the element, and another element may also be interposed between the element and the arbitrary element disposed on (or under) the element.

[0059] In addition, it will be understood that when a component is referred to as being “linked,” “coupled,” or “connected” to another component, the elements may be directly “coupled,” “linked” or “connected” to each other, or another component may be “interposed” between the components”.

[0060] Throughout the specification, when “A and/or B” is stated, it means A, B or A and B, unless otherwise stated. That is, “and/or” includes any or all combinations of a plurality of items enumerated. When “C to D” is stated, it means C or more and D or less, unless otherwise specified.

[0061] A battery pack according to one or more embodiments includes at least one battery module and a pack housing having an accommodation space in which the at least one battery module is accommodated.

[0062] The battery module may include a plurality of battery cells and a module housing. The battery cells may be accommodated inside the module housing in a stacked form (or stacked arrangement or configuration). Each battery cell may have a positive electrode terminal and a negative electrode terminal and may be a circular type, a prismatic type, or a pouch type according

to the shape of battery. In the present specification, a battery cell may also be referred to as a secondary battery, a battery, or a cell.

[0063] In the battery pack, one cell stack may constitute one module stacked in place of the battery module. The cell stack may be accommodated in an accommodation space of the pack housing or may be accommodated in an accommodation space partitioned by a frame, a partition wall, etc.

[0064] The terms used herein are intended to describe embodiments of the present disclosure, and are not intended to limit the present disclosure.

[0065] FIG. 1 illustrates an exploded perspective diagram showing a battery module **100** according to one or more embodiments of the present disclosure. For example, the battery module **100** may be a secondary battery module. Referring to FIG. 1, the battery module **100** may include a plurality of battery cells **110**, heat conducting members **120**, a first case **130**, a second case **140**, and a module cover **150**.

[0066] The battery cells **110** are chargeable and dischargeable, and may generate heat during charge/discharge. The configuration of the battery cells **110** will be described in more detail below with reference to FIG. 2 and FIG. 3.

[0067] In the first case **130**, a first internal space **132** may be formed to accommodate the battery cells **110**, and the battery cells **110** may be seated in the first internal space **132**. Further, in the first case **130**, a plurality of through regions may be formed to expose a portion of the surface of the battery cells **110**. The size/area of each of the through regions may be smaller than the size/area of the bottom surface of each of the battery cells. In addition, the first case **130** may have a plurality of heat exhaust holes (e.g., **432** in FIG. 4).

[0068] According to an embodiment, the first case **130** may include an insulating material. For example, the first case **130** may include at least one of polycarbonate, polyester, polyamide, and/or the like.

[0069] The second case **140** may have a second internal space **142** formed to accommodate at least a portion of the first case **130**. The first case **130** may be seated in the second internal space **142** of the second case **140** to be connected to (e.g., coupled to or attached to) the second case **140**. In addition, the second case **140** may have a plurality of heat exhaust holes (e.g., **452** in FIG. 4).

[0070] According to an embodiment, the second case **140** may include (e.g., may be made of) a metal having a high thermal conductivity. For example, the second case **140** may include at least one of aluminum (Al), an Al alloy, an Al oxide, steel, and/or the like.

[0071] The heat conducting members **120** may be disposed between the battery cells **110** and the second case **140**. For example, the heat conducting members may be heat conducting sheets, a heat conducting layer, or the like. In this case, in each of the heat conducting members **120**, a first surface may be in direct contact with a first surface of a corresponding battery cell (e.g., the bottom surface of the corresponding battery cell), and a second surface may be in direct contact with the second case **140**. As such, the heat conducting members **120** may transfer heat from the battery cells **110** to the second case **140**. The heat conducting members **120** may include (e.g., may be made of) a suitable material having a high thermal conductivity. For example, the heat conducting members **120** may be silicone sheets, acrylic sheets, urethane sheets, graphite sheets, and/or the like having a high thermal conductivity. For convenience, the heat conducting members **120** may be described in more detail hereinafter as being made of an insulating material.

[0072] In a state where the battery cells **110** are disposed inside the first case **130**, and the first case **130** is connected to (e.g., coupled to or attached to) the second case **140**, the module cover **150** may be connected to (e.g., coupled to or attached to) the first case **130** and the top portion of the second case **140**. The module cover **150** may have a plurality of through-holes **152** through which electrode terminals **112** disposed on the top surfaces of the battery cells **110** may extend.

[0073] The material of the module cover **150** may be a resin material. For example, the module cover **150** may include (e.g., may be made of) a thermoplastic resin, such as polycarbonate, polyethylene, polypropylene, acrylonitrile butadiene styrene (ABS) resin, polyamide, nylon, or

polyaryletherketone (PEEK), but the material of the module cover **150** is not limited to the foregoing examples. In addition, not only one type of material but a mixture of two or more types of materials may be used as the resin material.

[0074] In FIG. **1**, the battery cells **110** are shown as prismatic cells, but the present disclosure is not limited thereto. For example, the battery cells **110** may be other kinds of battery cells, such as cylindrical cells and pouch cells.

[0075] The first case **130** may act as an insulator and the second case **140** may act as a heat sink, thereby providing both insulation and heat dissipation for the battery cells **110**.

[0076] FIG. **2** illustrates a perspective view showing a secondary battery according to one or more embodiments of the present disclosure, and FIG. **3** is a cross-sectional view taken along the line II-II in FIG. **2**. Hereinafter, a secondary battery **200** may correspond to the battery cells **110** in FIG. **1**.

[0077] Referring to FIGS. **2** and **3**, the secondary battery **200** according to one or more embodiments of the present disclosure may include at least one electrode assembly **210** wound with a separator **213** as an insulator between the positive electrode **211** and the negative electrode **212**, a case **220** in which the electrode assembly **210** is received (or accommodated) therein, and a cap assembly **230** coupled to an opening of the case **220**.

[0078] The secondary battery **200** according to one or more embodiments illustrated in FIGS. **2** and **3** will now be described as an example of a prismatic lithium ion secondary battery. However, the present disclosure is not limited thereto, and suitable aspects, features and principles described herein may be applied to various other types of batteries, such as lithium polymer batteries and/or cylindrical batteries.

[0079] Each of the positive electrode **211** and the negative electrode **212** may include a current collector made of a thin metal foil having a coated portion on which an active material is coated and an uncoated portion **211a**, **212a** on which an active material is not coated.

[0080] The positive electrode **211** and the negative electrode **212** are wound after interposing the separator **213**, which is an insulator, therebetween. However, the present disclosure is not limited thereto, and the electrode assembly **210** may have a structure in which a positive electrode **211** and a negative electrode **212**, each made of a plurality of sheets, are alternately stacked with a separator interposed therebetween.

[0081] The case **220** may form the overall outer appearance of the secondary battery **200** and may be made of a conductive metal, such as aluminum, aluminum alloy, or nickel-plated steel. In addition, the case **220** may provide a space in which the electrode assembly **210** is accommodated.

[0082] The cap assembly **230** may include a cap plate **231** covering an opening in the case **220**, and the case **220** and the cap plate **231** may be made of a conductive material. The positive and negative electrode terminals **221** and **222** electrically connected to the positive electrode **211** and the negative electrode **212**, respectively, may be installed to penetrate (or extend through) the cap plate **231** and protrude outwardly therethrough.

[0083] In addition, outer peripheral surfaces (e.g., circumferential surfaces) of upper pillars of the positive and negative electrode terminals **221** and **222** protruding outwardly from the cap plate **231** may be threaded and may be fixed to the cap plate **231** by utilizing nuts.

[0084] However, the present disclosure is not limited thereto, and the positive and negative electrode terminals **221** and **222** may have a rivet structure and may be riveted or welded to the cap plate **231**.

[0085] In addition, the cap plate **231** may be made of a thin plate and may be coupled to the opening in the case **220**, and an electrolyte injection port **232** into which a sealing stopper **233** may be installed may be located (e.g., formed) in the cap plate **231**, and a vent portion **234** having a notch **234a** may be installed.

[0086] The positive and negative electrode terminals **221** and **222** may be electrically connected to current collectors including first and second current collectors **240** and **250** (hereinafter referred to as positive and negative current collectors) by being bonded or coupled (e.g., by welding) to the

positive uncoated portion **211a** and the negative electrode uncoated portion **212a**, respectively. [0087] For example, the positive and negative electrode terminals **221** and **222** may be coupled by welding to the positive and negative electrode current collectors **240** and **250**, respectively. However, the present disclosure is not limited thereto, and the positive and negative electrode terminals **221** and **222** and the positive and negative electrode current collectors **240** and **250** may be integrally formed in one or more embodiments.

[0088] In addition, an insulation member may be installed between the electrode assembly **210** and the cap plate **231**. The insulation member may include first and second lower insulation members **260** and **270**, and each of the first and second lower insulation members **260** and **270** may also have a portion located between the electrode assembly **210** and the case **220**.

[0089] In addition, according to one or more embodiments of the present disclosure, one end of a separation member may face one side of the electrode assembly **210** and may be installed between the insulation member and the positive or negative electrode terminals **221** and **222**.

[0090] In one or more embodiments, the separation member may include first and second separation members **280** and **290**.

[0091] In such an embodiment, first ends of the first and second separation members **280** and **290** installed to face one side of the electrode assembly **210** may be respectively installed between the first and second lower insulation members **260** and **270** and the positive and negative electrode terminals **221** and **222**.

[0092] Accordingly, the positive and negative electrode terminals **221** and **222**, which may be coupled by welding to the positive and negative electrode current collectors **240** and **250**, may be coupled to first ends of the first and second lower insulation members **260** and **270** and the first and second separation members **280** and **290**.

[0093] FIG. **4** illustrates a perspective view showing a first case **410** and a second case **420** according to one or more embodiments of the present disclosure. The first case **410** in FIG. **4** may correspond to the first case **130** described above with reference to FIG. **1**, and the second case **420** in FIG. **4** may correspond to the second case **140** described above with reference to FIG. **1**.

[0094] According to an embodiment, the first case **410** may be provided with a first internal space **412** for accommodating at least one battery cell. A plurality of through regions **414** may be formed in the bottom surface of the first case **410** within the first internal space **412**.

[0095] According to an embodiment, the area of each of the through regions **414** may be smaller than the area of a first surface (e.g., a bottom side) of the corresponding battery cell. In addition or in another example, the length of the long sides of each of the through regions **414** may be shorter than the length of the long sides of the first surfaces of the battery cells. In addition or in another example, the length of the short sides of the through regions **414** may be shorter than the length of the short sides of the first surface of the battery cells. As such, the battery cells may be fixed within the first internal space **412** of the first case **410**.

[0096] According to an embodiment, the number of the through regions **414** may be equal to the number of the battery cells accommodated in the first internal space **412**. In some embodiments, the number of the through regions **414** may be less than or greater than the number of the battery cells accommodated in the first internal space **412**.

[0097] According to an embodiment, the first case **410** may include a plurality of heat exhaust holes **432**. For example, the heat exhaust holes **432** may be provided on side surfaces and/or the bottom surface of the first case **410**. The positions of the heat exhaust holes **432** of the first case **410** may correspond to the positions of the heat exhaust holes **452** of the second case **420**. The heat exhaust holes **432** of the first case **410** are shown as having a shape of elliptical through-holes, but the present disclosure is not limited thereto, and the heat exhaust holes **432** may have any suitable shape.

[0098] According to an embodiment, the first case **410** may be provided with compartment walls **434** to define regions in which the battery cells are to be disposed. For example, the compartment

walls **434** may be formed by protruding from the bottom surface of the first case **410**. In an embodiment, the compartment walls **434** may be configured such that one battery cell is disposed in each of the regions defined by the compartment walls **434**. In another embodiment, the compartment walls **434** may be configured such that two or more battery cells are disposed in each of the regions defined by the compartment walls **434**.

[0099] According to an embodiment, the first case **410** may be provided with guide protrusions **436** to guide the mounting of the battery cells. For example, the guide protrusions **436** may be provided on the side surfaces of the first case **410** and/or on opposite sides of the compartment walls **434**. In addition, the guide protrusions **436** may structurally reinforce the first case **410**. The first case **410** may be provided with the through regions **414** at (e.g., in or on) which the heat conducting members are to be disposed, and a plurality of through-holes for forming the heat exhaust holes **432**, and thus, structural reinforcement may be desired. Accordingly, the guide protrusions **436** may be formed on the side surfaces of the first case **410** and/or on the opposite sides of the compartment walls **434** to protrude outwardly, thereby structurally reinforcing the first case **410**.

[0100] In addition, support protrusions **438** may be provided on the first case **410** to structurally reinforce the first case **410**. For example, the support protrusions **438** may be provided by protruding outwardly in a grid-like pattern from the side surfaces of the first case **410**.

[0101] According to an embodiment, a space **440** may be formed in the first case **410** in which a battery management module (e.g., a battery management system) is disposed. Further, the first case **410** may include an engagement member (e.g., a coupling member) **442** formed for connecting (e.g., coupling or attaching) with the battery management module. The number, the shape, and the positions of the engagement members **442** for connecting (e.g., coupling or attaching) with the battery management module are not limited to the example shown in FIG. 4, and they may be provided in different numbers, shapes, and positions as needed or desired.

[0102] According to an embodiment, the second case **420** may be provided with a second internal space **422** that may accommodate at least a portion of the first case **410**. The first case **410** and the second case **420** may be connected (e.g., coupled or attached) together, such that the first case **410** is seated in the second internal space **422** provided in the second case **420**.

[0103] In more detail, the first case **410** and the second case **420** may each be provided with a coupling member, and may be coupled/fixed together using the coupling members. In an embodiment, the first case **410** and the second case **420** may include engagement holes **446** and **454** for screw engagement. For example, the first case **410** and the second case **420** may have the engagement holes **446** and **454** for allowing screws to be inserted at corresponding positions in the coupling.

[0104] In an embodiment, one or more engagement protrusions may be provided on an outer surface of the first case **410** (e.g., on a lower portion of the first case **410**), and engagement grooves **458** may be provided on an inner surface of the second case **420** at positions corresponding to the engagement protrusions. In this case, the engagement protrusions of the first case **410** may be inserted into the engagement grooves **458** of the second case **420**, thereby allowing the first case **410** and the second case **420** to be connected (e.g., coupled or attached) together.

[0105] In an embodiment, the first case **410** and the second case **420** may be coupled/fixed together in a hook engagement manner. For example, the engagement protrusions of the first case **410** may include a clasp. In this case, the first case **410** and the second case **420** may be coupled/fixed together, such that the clasps of the engagement protrusions of the first case **410** are inserted into the engagement grooves **458** of the second case **420**. The manner in which the first case **410** and second case **420** are connected (e.g., coupled or attached) together is not limited to the examples described above, and they may be connected (e.g., coupled or attached) together in various suitable manners.

[0106] In an embodiment, protrusions may be provided on an outer surface of the first case **410** (e.g., on a lower portion of the first case **410**) to guide engagement positions between the first case

410 and the second case **420**. On an inner surface of the second case **420**, through-holes **456** may be formed at positions corresponding to the protrusion. The through-holes **456** of the second case **420** may have a shape corresponding to the protrusions of the first case **410**. In this case, the protrusions of the first case **410** may be inserted into the through-holes **456** of the second case **420** to guide the engagement positions of the first case **410** and the second case **420**.

[0107] According to an embodiment, the second case **420** may include a plurality of heat exhaust holes **452**. For example, the heat exhaust holes **452** may be provided on side surfaces and/or the bottom surface of the second case **420**. The positions of the heat exhaust holes **452** of the second case **420** may correspond to the positions of the heat exhaust holes **432** of the first case **410**. The heat exhaust holes **452** of the second case **420** are shown as having a shape of elliptical through-holes, but the present disclosure is not limited thereto, and the heat exhaust holes **452** may have various suitable shapes.

[0108] According to an embodiment, convex portions **460** may be provided on the second case **420** to structurally reinforce the second case **420**. For example, the convex portions **460** may be provided so that portions of the inner surfaces of the second case **420** protrude inwardly, and portions of the outer surfaces of the second case **420** protrude outwardly.

[0109] FIG. 5 illustrates a perspective diagram showing a coupled state of the first case **410** and the second case **420** according to one or more embodiments of the present disclosure. FIG. 6 illustrates a plan diagram showing the coupled state of the first case **410** and the second case **420** according to one or more embodiments of the present disclosure.

[0110] Referring to FIG. 5 and FIG. 6, when the first case **410** and the second case **420** are connected (e.g., coupled or attached) together, portions of the second case **420** may be exposed to the first internal space **412** through the through regions **414**. In FIG. 5 and FIG. 6, eight rectangular through regions **414** having the same size as each other are shown as being formed in the first case **410**. However, the shape, the size, and the number of the through regions **414** are not limited thereto.

[0111] FIG. 7 illustrates a state where heat conducting members **710** are disposed in a plurality of through regions according to one or more embodiments of the present disclosure. As shown in FIG. 7, the heat conducting members **710** may be disposed within the through regions **414** (e.g., see FIG. 6) formed in the first case **410**. Herein, the number of the through regions **414** may be the same as the number of the heat conducting sheets of the heat conducting members **710**. Further, the through regions may correspond to the battery cells.

[0112] In an embodiment, the length of the long sides of the heat conducting members **710** may be shorter than or equal to the length of the long sides of the through regions **414**, and the length of the short sides of the heat conducting members **710** may be shorter than or equal to the length of the short sides of the through regions **414**. Accordingly, each of the heat conducting members **710** may be accommodated within a corresponding one of the through regions **414**, and may be in direct contact with the second case **420**.

[0113] According to an embodiment, the thickness of the heat conducting members **710** may be thicker than the thickness of the first case **410** in which the through regions **414** are provided. Accordingly, the second surfaces opposite the first surfaces of the heat conducting members **710** that are in direct contact with the second case **420** may be in direct contact with the first surfaces (e.g., the bottom sides) of the battery cells.

[0114] FIG. 8 illustrates an example of a battery module **800** in which a plurality of battery cells **810** are inserted in the first case **410** in a state where the first case **410** and the second case **420** are coupled together according to one or more embodiments of the present disclosure. The A-A' cross-sectional view of FIG. 8 illustrates the cross-section taken along the line A-A' of the region B of the battery module **800** shown in FIG. 8. The battery cells **810** shown in FIG. 8 may correspond to the battery cells **110** described above with reference to FIG. 1.

[0115] Referring to FIG. 8, the battery cells **810** may be seated in a first internal space defined in

the first case **410** in which a plurality of heat conducting members **820** are disposed in the through regions. Accordingly, first surfaces of the heat conducting members **820** may be in direct contact with portions of the surfaces of the battery cells **810** that are exposed through the through regions, and second surfaces of the heat conducting members **820** may be in contact with the second case **420**. For example, from among the heat conducting members **820**, a first heat conducting member **822** may be disposed in a first through region **832**, a first surface of the first heat conducting member **822** may be in direct contact with a surface of a first battery cell **812** exposed through the first through region **832**, and a second surface of the first heat conducting member **822** may be in contact with the second case **420**. Further, from among the heat conducting members **820**, a second heat conducting member **824** may be disposed in a second through region **834**, a first surface of the second heat conducting member **824** may be in direct contact with an exposed surface of a second battery cell **814** through the second through region **834**, and a second surface of the second heat conducting member **824** may be in contact with the second case **420**.

[0116] As described above, heat generated by the battery cells **810** may be dissipated outwardly through the second case **420** by means of the heat conducting members **820**. In more detail, as the battery cells **810** are surrounded (e.g., around peripheries thereof) by the first case **410** including an insulating material having a low thermal conductivity, heat may be generated and temperature may rise in the first internal space (e.g., **132** in FIG. **1**) in which the battery cells **810** are positioned. However, according to one or more embodiments of the present disclosure, the temperature of the battery cells **810** seated in the first case **410** may be lowered, because heat from the battery cells **810** may be transferred to and dissipated outwardly through the heat conducting members **820** and the second case **420**.

[0117] In addition, as the through regions **832** and **834** are formed in the first case **410**, and the heat conducting members **820** are disposed in the through regions **832** and **834**, a larger area of the heat conducting members **820** may be in contact with the battery cells **810**. Further, as the area of each of the through regions **832** and **834** and each of the heat conducting members **820** are increased, a larger area of single heat conducting members **822** and **824** may be in contact with single battery cells. With the increase in the total area over which the battery cells **810** are in contact with the heat conducting members **820**, the heat dissipation effect of the battery cells may be improved.

[0118] Further, a plurality of holes having various suitable shapes may be provided in the first case **410** and the second case **420**. Through the holes, heat from the battery cells **810** may be dissipated outwardly.

[0119] FIG. **9** illustrates through regions having various shapes according to one or more embodiments of the present disclosure. First to third examples **910** to **930** in FIG. **9** illustrate through regions **912**, **922**, and **932**, respectively, corresponding to two or more battery cells. In order to obtain a sufficient heat dissipation effect, the area of the through regions **912**, **922**, and **932** may be set so that the area where the heat conducting members and the battery cells are in contact with each other is at least a suitable sized area (e.g., a predetermined area). As the number of through regions is less than the number of battery cells, the process of forming the through regions in the first case and the process of preparing and arranging the heat conducting members may be simplified.

[0120] Referring to the first example **910**, a plurality of through regions **912** having a wider transverse width and a narrower longitudinal width may be formed in the first case. In this case, first surfaces (e.g., longitudinal sides) of the through regions **912** may be shorter than first surfaces of the battery cells, and second surfaces (e.g., transverse sides) of the through regions **912** may be longer than second surfaces of the battery cells. Because the first surfaces of the through regions **912** are shorter than the first surfaces of the battery cells, the battery cells may be seated without passing through the through regions **912**. In the first example **910**, the respective through regions may correspond to four battery cells. In this case, portions of the surfaces of the cells may be exposed through the individual through regions. In the first example **910**, portions of the surface of

the four battery cells are shown as exposed through a pair of through regions **912**. The area of contact between the battery cells and the heat conducting members disposed in the through regions **912** may be equal to or greater than a predetermined area.

[0121] Referring to the second example **920**, the respective through regions **922** may correspond to four battery cells. In this case, surface portions of a plurality of cells may be exposed through a single through region **922**. In the second example **920**, portions of the surfaces of the four battery cells are shown as exposed through the single through region **922**. The area of contact between the battery cells and the heat conducting members disposed in the through regions **922** may be equal to or greater than a predetermined area.

[0122] Referring to the third example **930**, first surfaces (e.g., longitudinal sides) of the through regions **932** may be longer than first surfaces of the battery cells, and second surfaces (e.g., transverse sides) of the through regions **932** may be shorter than second surfaces of the battery cells. Because the second surfaces of the through regions **932** are shorter than the second surfaces of the battery cells, the battery cells may be seated without passing through the through regions **932**. In the third example **930**, the respective through regions **932** may correspond to two battery cells. In this case, portions of the surfaces of the cells may be exposed through the respective through regions **932**. In the third example **930**, portions of the surfaces of the two battery cells are shown as exposed through a single through region **932**. The area of contact between the battery cells and the heat conducting members disposed in the through regions **932** may be equal to or more than a predetermined area.

[0123] In the first example **910**, the second example **920**, and the third example **930**, a plurality of heat conducting members corresponding to the sizes of the respective corresponding through regions may be disposed in the through regions.

[0124] In addition, a plurality of through regions may correspond to a single battery cell. In order to obtain a sufficient heat dissipation effect even in this case, the area of the through regions may be set so that the area in contact between the heat conducting members and the battery cells is equal to or more than a predetermined area. In an embodiment, a greater number of through regions than the number of battery cells may be provided in the first case. For example, based on the number of battery cells, through regions corresponding to a multiple of n (where n is a real number exceeding 1) may be provided in the first case. In this case, a surface of a single battery cell may be exposed through a plurality of through regions, and each of the through regions may have a smaller size compared to that of the example described above with reference to FIG. 6. For example, when a surface of a single battery cell is exposed through a pair of through regions, the total area of the pair of through regions may be smaller than the area of a first surface (e.g., the bottom side) of the battery cell.

[0125] In the embodiments described above, the through regions are illustrated as having a rectangular shape, but the present disclosure is not limited thereto, and through regions having various suitable shapes may be provided in the first case. For example, the first case may be provided with through regions having a suitable shape, such as a polygon or a circle, and having a suitable size so that the battery cells may be seated without passing through. Further, heat conducting members having a suitable shape, such as a polygon or a circle, may be disposed in the through regions. The shape of the through regions may be determined differently depending on the battery type (e.g., a prismatic type, a cylindrical type, or a pouch type).

[0126] FIGS. **10** and **11** illustrate a battery pack according to one or more embodiments of the present disclosure.

[0127] The battery pack may include a plurality of battery modules **50** and a housing **10** for accommodating the battery modules **50**. For example, the housing **10** may include first and second housings **11** and **12** coupled in opposite directions through the battery modules **50**. The battery modules **50** may be electrically connected to each other by using a bus bar **51**, and the battery modules **50** may be electrically connected to each other in a series/parallel or series-parallel mixed

method, thereby obtaining desired (e.g., required) electrical output.

[0128] The battery pack may include a high-voltage connector **60** exposed to the outside of the housing **10**. For example, the high-voltage connector **60** may be provided on a first housing **11** forming the upper portion of the housing **10**. The battery pack may provide a drive power source for vehicles, and the high-voltage connector **60** may form an output terminal for supplying the drive power source.

[0129] FIGS. **12** and **13** illustrate a vehicle body and vehicle body parts having a battery pack according to one or more embodiments of the present disclosure.

[0130] In FIG. **12**, a battery pack **91** may include a battery pack cover **23**, which is a part of a vehicle underbody **92**, and a pack frame **20** disposed under the vehicle underbody **92**. The pack frame **20** and the battery pack cover **23** may be integrally formed with a vehicle floor **82**.

[0131] The vehicle underbody **92** separates the inside and outside of a vehicle, and the pack frame **20** may be disposed outside the vehicle.

[0132] FIG. **13** illustrates a schematic side view of a vehicle according to one or more embodiments of the present disclosure.

[0133] A vehicle **1300** may be formed by combining additional parts, such as a hood **97** in front of the vehicle and fenders **98** respectively located in the front and rear of the vehicle to a vehicle body **99**.

[0134] The vehicle **1300** may further include a vehicle floor **82**, which is one of the vehicle body parts **90** including the battery pack **91** including the pack frame **20** and the battery pack cover **23**.

[0135] FIG. **14** illustrates a flowchart showing a method **1400** of assembling a secondary battery module according to one or more embodiments of the present disclosure.

[0136] First, the assembling method **1400** may start, and a first case may be prepared (**S1410**) for accommodating battery cells, and may be provided with a plurality of through regions through which portions of the surfaces of the battery cells are exposed. Here, the first case may include an insulating material. Further, each of the through regions may correspond to a single battery cell. In another example, each of the through regions may correspond to two or more battery cells. In another example or in addition, each of the battery cells may correspond to two or more through regions.

[0137] According to an embodiment, the through regions may be provided in the first case, such that the area of the through regions is smaller than the area of first surfaces of the battery cells. According to some embodiments, the through regions may be provided in the first case, such that the length of long sides of the through regions is shorter than the length of long sides of first surfaces of the battery cells, and the length of short sides of the through regions is shorter than the length of short sides of the first surfaces of the battery cells.

[0138] A second case may be prepared (**S1420**) to have an internal space to receive at least a portion of the first case. Here, the second case may include (e.g., may be made of) a metal having a high thermal conductivity.

[0139] The first case and the second case may be coupled together (**S1430**), such that the first case is received in the internal space provided in the second case.

[0140] Heat conducting sheets (e.g., heat conducting members) may be disposed in each of the through regions of the first case (**S1440**). Here, the heat conducting members may include (e.g., may be made of) a suitable material having a high thermal conductivity.

[0141] The battery cells may be seated in the first case (**S1450**), so that the battery cells are disposed in the through regions, and the method **1400** may end.

[0142] According to one or more embodiments, the battery cells may be seated in the first case, such that first surfaces of the heat conducting members are in direct contact with portions of the surfaces of the battery cells exposed through the through regions. In this case, second surfaces of the heat conducting members may be in contact with the second case.

[0143] Although the present disclosure has been described with reference to embodiments and

drawings illustrating aspects thereof, the present disclosure is not limited thereto. Various modifications and variations can be made by a person skilled in the art to which the present disclosure belongs within the scope of the technical spirit of the present disclosure and the claims and their equivalents, below.

DESCRIPTION OF REFERENCE SYMBOLS

[0144] **100**: battery module [0145] **110**: battery cell [0146] **120**: heat conducting member [0147] **130**: first case [0148] **140**: second case

Claims

1. A battery module comprising: a first battery cell; a first case configured to accommodate the first battery cell, and having a first through region to expose a portion of a surface of the first battery cell therethrough; a second case configured to accommodate at least a portion of the first case; and a first heat conducting member between the first battery cell and the second case.
2. The battery module as claimed in claim 1, wherein a first surface of the first heat conducting member is in direct contact with a portion of a surface of the first battery cell exposed through the first through region, and a second surface of the first heat conducting member is in contact with the second case.
3. The battery module as claimed in claim 1, wherein the first heat conducting member is located in the first through region of the first case.
4. The battery module as claimed in claim 1, wherein an area of the first through region is smaller than an area of a first surface of the first battery cell.
5. The battery module as claimed in claim 1, wherein a length of a long side of the first through region is shorter than a length of a long side of a first surface of the first battery cell, and wherein a length of a short side of the first through region is shorter than a length of a short side of the first surface of the first battery cell.
6. The battery module as claimed in claim 1, wherein the first case is configured to accommodate a plurality of battery cells comprising the first battery cell, wherein the first case has a plurality of through regions comprising the first through region, wherein each of the through regions corresponds to a single battery cell from among the plurality of battery cells, and wherein a plurality of heat conducting members comprising the first heat conducting member is located in the through regions, respectively.
7. The battery module as claimed in claim 1, wherein the first case is configured to accommodate a plurality of battery cells comprising the first battery cell, wherein the first case has a plurality of through regions comprising the first through region, wherein each of the through regions corresponds to two or more battery cells from among the plurality of battery cells, and wherein a plurality of heat conducting members comprising the first heat conducting member is located in the through regions, respectively.
8. The battery module as claimed in claim 1, wherein a thickness of the first heat conducting member is thicker than a thickness of the first case having the first through region.
9. The battery module as claimed in claim 1, wherein the first case comprises an insulating material.
10. The battery module as claimed in claim 1, wherein the second case comprises a metal.
11. The battery module as claimed in claim 1, further comprising: a second battery cell; and a second heat conducting member between the second battery cell and the second case, wherein the second case has a second through region to expose a portion of a surface of the second battery cell therethrough.
12. A method of assembling a battery module, the method comprising: preparing a first case for accommodating a battery cell, and having a through region to expose a portion of a surface of the battery cell therethrough; preparing a second case having an internal space for accommodating at

least a portion of the first case; coupling the first case and the second case together so that the first case is seated in the internal space of the second case; disposing a heat conducting member in the through region of the first case; and seating the battery cell in the first case to be disposed on the through region.

13. The method as claimed in claim 12, wherein, in the seating of the battery cell in the first case, a first surface of the heat conducting member is in direct contact with a portion of a surface of the battery cell exposed through the through region, and a second surface of the heat conducting member is in contact with the second case.

14. The method as claimed in claim 12, wherein, in the preparing of the first case, the through region is formed in the first case so that an area of the through region is smaller than an area of a first surface of the battery cell.

15. The method as claimed in claim 12, wherein, in the preparing of the first case, the through region is formed in the first case so that a length of a long side of the through region is shorter than a length of a long side of a first surface of the battery cell, and a length of a short side of the through region is shorter than a length of a short side of the first surface of the battery cell.

16. The method as claimed in claim 12, wherein the first case is configured to accommodate a plurality of battery cells comprising the battery cell, wherein the first case has a plurality of through regions comprising the through region, wherein each of the through regions corresponds to a single battery cell from among the plurality of battery cells, and wherein a plurality of heat conducting members comprising the heat conducting member is disposed in the through regions, respectively.

17. The method as claimed in claim 12, wherein the first case is configured to accommodate a plurality of battery cells comprising the battery cell, wherein the first case has a plurality of through regions comprising the through region, wherein each of the through regions corresponds to two or more battery cells from among the plurality of battery cells, and wherein a plurality of heat conducting members comprising the heat conducting member is disposed in the through regions, respectively.

18. The method as claimed in claim 12, wherein a thickness of the heat conducting member is thicker than a thickness of the first case having the through region.

19. The method as claimed in claim 12, wherein the first case comprises an insulating material.

20. The method as claimed in claim 12, wherein the second case comprises a metal.
